

Experiment - 12

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(A)

Design a trigger which whenever there is an insertion on student table then, the currently inserted or deleted row should be printed as it as on the output console window.

Solution:

Create or replace function fn_student_audit() returns trigger

Language plpgsql as

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begin

 If tg_op = 'insert' then

 Raise notice 'inserted row -> id: %, name: %, age: %, class:

%',

 new.id, new.name, new.age, new.class; return new;

 Elsif tg_op = 'delete' then

 Raise notice 'deleted row -> id: %, name: %, age: %, class:

%', Old.id, old.name, old.age, old.class; return old;

 End if; return

 null;

End;

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Create trigger trg_student_audit after insert or delete
On student for each
row
Execute function fn_student_audit();

Insert into student(name, age, class) values('john', 17, 11); delete from student
Where name='jack';

(B)

Design a POSTGRESQL triggers that:

Whenever a new employee is inserted in tbl_employee, a record should be added to tbl_employee_audit like:

"Employee name <emp_name> has been added at <current_time>"

Whenever an employee is deleted from tbl_employee, a record should be added to tbl_employee_audit like:

"Employee name <emp_name> has been deleted at

<current_time>" The solution must use PostgreSQL triggers.

Solution:

Create or replace function audit_employee_changes() returns trigger

Language plpgsql as

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begin

 If tg_op = 'insert' then

 Insert into tbl_employee_audit(message)

 Values ('employee name ' || new.emp_name || ' has been added at ' || now());

 Return new;

 Elsif tg_op = 'delete' then

 Insert into tbl_employee_audit(message)

 Values ('employee name ' || old.emp_name || ' has been deleted at ' || now());

 Return old;

End if;

Return null; end;

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Create trigger trg_employee_audit after insert or delete

On tbl_employee for

each row

Execute function audit_employee_changes();

Insert into tbl_employee(emp_name, emp_salary) values
('alan', 50000);

Delete from tbl_employee where

emp_name='alan';

Select * from tbl_employee_audit;